

# Heart Disease Prediction using various Logistic Regression, Random Forest Classifier and Support Vector Classifier

Semil Halani  
210147012  
MSc Computer Science

**Abstract**—This research paper studies the various factors that are related to heart diseases and then is used to predict if there can be a futuristic scenario where the person with specific features and qualities might suffer from a heart disease or not. The dataset used in this study is a subset of a larger database with more data irrelevant to our research and so, this dataset was created and is available from a public repository. It describes various features related to developing a heart disease in a person. The data is then analysed, preprocessed and scaled for the use of our research so that it performs the best and fits well with our models. Three Machine Learning classification models are used here, namely Logistic Regression, Random Forest Classifier and Support Vector Classifier and the results are then compared using various comparison metrics.

**Keywords**—*heart disease prediction, machine learning, heart failure, data analysis, logistic regression, random forest classification, support vector classification*

## I. INTRODUCTION

In this digital fast-paced world, time and health, both seem to fly away if not taken care of meticulously. With the advent of technology, there has been developments in almost every field of life. It feels as if life revolved around technology and also, technology has surrounded life as well. With exponential increase of use of technologies in our daily life and even at industrial and research levels, there has been an abundant accumulation of data in the past few years. This data is being used for various good or bad motives like, marketing, businesses, development and even as bad as something like hacking. Recently, there have been lots and lots of developments in this field of data

analytics and use of machine learning for the development of the society. It is evident from the number of researches ongoing in this field that the world is moving towards a new bright sunny day, a day with light (knowledge and development) where there would be little to no causes of illness or danger or any other negative circumstances.

Even Machine Learning is a booming field right now. The invention of robots and self-driving cars as described by Soni et al. (2021)<sup>[1]</sup>, acts like behaviour cloning which were only a fiction of books and movies are becoming reality, that too in our daily life instances. This paper reflects on the positive use of data accumulated by a research organization. The dataset used in this paper is a subset of a larger dataset with a lot of irrelevant data available of the website of UC Irvine Machine Learning Repository. There are numerous amount of machine learning techniques like Linear Regression, Logistic Regression, Decision Tree Classifier, Support Vector Machines, Random Forest Classifier, and Deep Learning models like Feedforward Neural Network, Convolutional Neural Network, Recurrent Neural Network, etc. This problem at hand regarding the disease prediction can be considered as a classification problem as either there will be a suffering of disease in the future or there will be a healthy life without any heart diseases. Hence, in this paper, we would be using three Machine Learning classification models namely, Logistic Regression, Random Forest Classifier, and Support Vector Classifier.

## II. LITERATURE REVIEW

Disease prediction has been a significantly favourite part of the advanced Machine Learning researches that have been in trend since a long time. There have been plenty of searches for diseases like malaria, cancer, brain tumour, etc. Cruz et al. (2006)<sup>[2]</sup> describes about a hybrid classifier of

EFuNN classifier and Bayesian classifier with an accuracy of 87.5%. Similarly, Turkki et al. (2019)<sup>[3]</sup> classified tumour tissues images into low or high digital risk score groups with a 95% accuracy.

Furthermore, research is done for other diseases as well. Kalipe et al. (2018)<sup>[4]</sup> discuss about various machine learning and deep learning approaches like KNN, Naive Bayes, and Extreme Gradient Boost etc. for predicting Malarial Outbreak. Furthermore, according to the study by Sramka et al. (2019)<sup>[5]</sup>, the often regarded as the most accurate formula for clinical refractive results of cataract surgery, Support Vector Machine Regression Model and Multilayer Neural Network Ensemble model beat that formula with a better result.

### III. DATASET

The dataset used here is directly available publicly and is a processed subset of a larger dataset published by the UC Irvine Machine Learning Repository. It is available at <https://archive.ics.uci.edu/ml/datasets/Heart+Disease>.

This dataset has around 918 rows and 12 columns. It provides details about people and specific characteristics related to them and states if they suffer from a heart disease or not. Using this data, it can be studied and analysed to prepare the dataset such that it can be used to fit our model and predict future cases and prevent them at an early point of time. Following are the various columns which are used in the dataset and their descriptions as to what they represent:

|   | Attribute Title | Attribute Description                                                             |
|---|-----------------|-----------------------------------------------------------------------------------|
| 1 | Age             | Age of the person under observation                                               |
| 2 | Sex             | Sex of the person under observation namely Male or Female in the current dataset. |
| 3 | ChestPainType   | The type of chest pain being suffered by the person under observation.            |

|    |                |                                                                          |
|----|----------------|--------------------------------------------------------------------------|
| 4  | Resting BP     | Value for the Resting Blood Pressure for the person under consideration. |
| 5  | Cholesterol    | Value for the Serum Cholesterol                                          |
| 6  | FastingBS      | Fasting Blood Sugar, whether > 120 mg/dl                                 |
| 7  | RestingECG     | Value for the results of resting Electrocardiography with                |
| 8  | MaxHR          | Maximum Heart rate observed for the person under observation.            |
| 9  | ExerciseAngina | Angina induced by Exercise                                               |
| 10 | Oldpeak        | ST depression induced by exercise relative to rest                       |
| 11 | ST_Slope       | the slope of the peak exercise ST segment                                |
| 12 | HeartDisease   | Chances of a heart attack/heart disease.                                 |

As described in the table above, all the features are mentioned there. Furthermore, there are four different types of chest pain under consideration. They are: typical angina, atypical angina, non-anginal pain and asymptomatic chest pain. Similarly, while recording data for the FastingBS field, exact data is not been under consideration but instead, it has Boolean datatype and is field as 1 if the fasting blood sugar is greater than 120 mg/dl and is recorded 0 otherwise. On the other hand, Resting ECG also has a similar datatype compared to Cholesterol, i.e. it has three distinct values and is saved accordingly. The resting Electrocardiographic results can either be normal, have ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) or show probable or definite left ventricular hypertrophy by Estes' criteria. Exercise Induced Angina field is similar to that of FastingBS, i.e. binary, as 1 if the angina is induced by exercise and

0 if not. In a similar pattern, the ST\_Slope has three distinct values as well, namely upsloping, flat and downsloping.

#### IV. DATA EXPLORATION

One of the most significant processes that one needs to undergo before using a dataset is initially understanding the dataset. Only after understanding a dataset, one can conclude to what needs to be done on the dataset and how it can be used further. Without understanding a dataset, there can be no proper pre-processing. This further results in a poor trained model. Thus, understanding a dataset is one of the most significant steps for data analysis.

Our dataset contains a few categorical variables as well. These categorical variables can be converted into distinct numbers and then, it would be very easy to use the correlation function on the entire dataset. So, all the categorical variables are converted to distinct numerical variables and the correlation function is applied then.

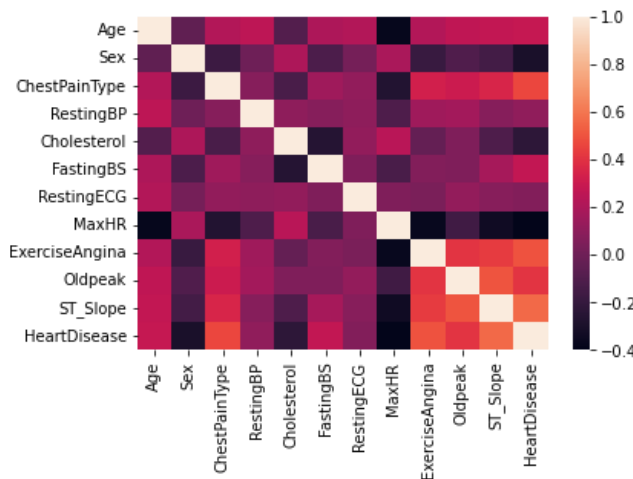


Figure 1. Correlation Heatmap

Through correlation we see that most of the columns are not interdependent and so, we cannot remove any more columns. We can conclude saying that all current columns would be our features.

Furthermore, the dataset is very evenly distributed as seen in the charts below:

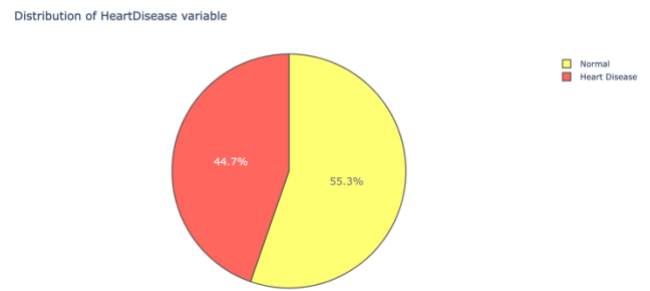


Figure 2. General Data Distribution

On further analysis of the data, we can see how the data is distributed among various features like age, sex, chest pain type, etc.

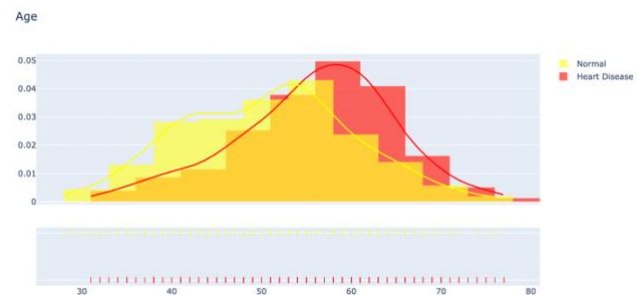


Figure 3. Data Distribution over Age

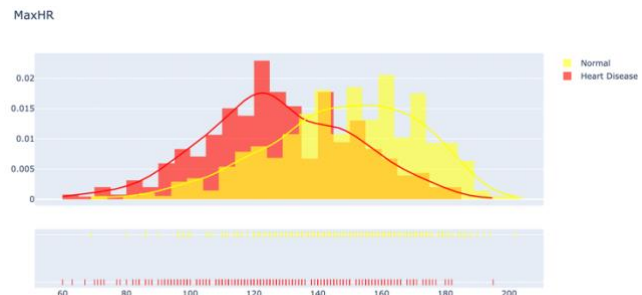


Figure 4. Data Distribution over Maximum Heart Rate achieved

Similarly, the following results are found upon exploring individual features one-vs-one for our data as shown below:

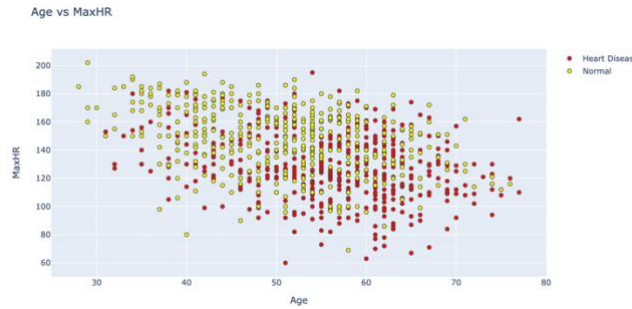


Figure 5. Data Distribution over Age vs Maximum Heart Rate achieved

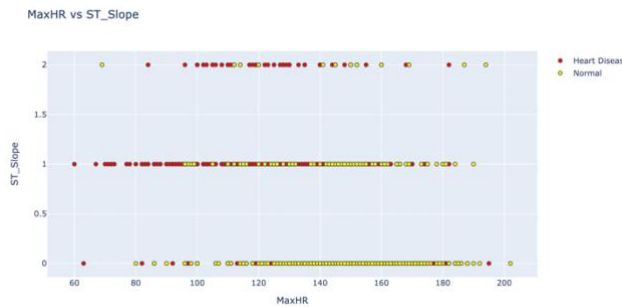


Figure 6. Data Distribution over Maximum Heart Rate vs ST\_Slope

## V. DATA PREPROCESSING

After analysing the data, now one can conclude to what changes are needed if any. Here, our dataset has only 12 columns of which most of them are uncorrelated. Hence, we cannot reduce any more columns else it would negatively affect our research as we might not have significant data for training the models.

Before starting with any processes, initially, it has to be checked if there are any missing or null values in our dataset. If we find any, then they can be replaced with a mean value of the column of if we have a huge dataset, those records with missing attribute values can be dropped as well.

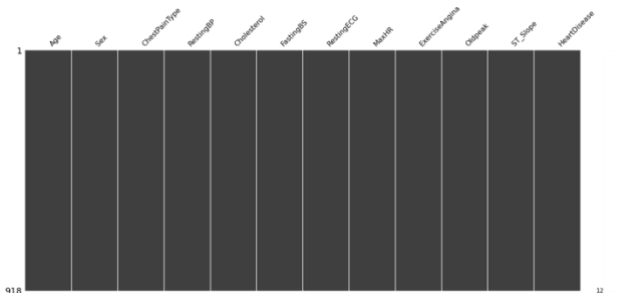


Figure 7. Checking Missing Values in our dataset

As we do not have any missing values or null values in our dataset, we are good to go to the next stage of the process. Now, on checking the description of our dataset,

|       | Age        | Sex        | ChestPainType | RestingBP  | Cholesterol | FastingBS  | RestingECG | MaxHR      | ExerciseAngina | Oldpeak    | ST_Slope   | HeartDisease |
|-------|------------|------------|---------------|------------|-------------|------------|------------|------------|----------------|------------|------------|--------------|
| count | 918.000000 | 918.000000 | 918.000000    | 918.000000 | 918.000000  | 918.000000 | 918.000000 | 918.000000 | 918.000000     | 918.000000 | 918.000000 | 918.000000   |
| mean  | 53.510893  | 0.210240   | 1.452070      | 132.396514 | 198.799564  | 0.233115   | 0.603486   | 136.809398 | 0.404139       | 0.887364   | 0.638344   | 0.553        |
| std   | 9.432817   | 0.407701   | 0.851832      | 18.514154  | 109.384145  | 0.423046   | 0.805968   | 25.480334  | 0.490992       | 1.066570   | 0.607056   | 0.497        |
| min   | 28.000000  | 0.000000   | 0.000000      | 0.000000   | 0.000000    | 0.000000   | 0.000000   | 60.000000  | 0.000000       | -2.600000  | 0.000000   | 0.000000     |
| 25%   | 47.000000  | 0.000000   | 1.000000      | 120.000000 | 173.250000  | 0.000000   | 0.000000   | 120.000000 | 0.000000       | 0.000000   | 0.000000   | 0.000000     |
| 50%   | 54.000000  | 0.000000   | 2.000000      | 130.000000 | 223.000000  | 0.000000   | 0.000000   | 138.000000 | 0.000000       | 0.600000   | 1.000000   | 1.000000     |
| 75%   | 60.000000  | 0.000000   | 2.000000      | 140.000000 | 267.000000  | 0.000000   | 1.000000   | 156.000000 | 1.000000       | 1.500000   | 1.000000   | 1.000000     |
| max   | 77.000000  | 1.000000   | 3.000000      | 200.000000 | 603.000000  | 1.000000   | 2.000000   | 202.000000 | 1.000000       | 6.200000   | 2.000000   | 1.000000     |

Figure 8. Dataset Value Description

one can say that the values pertaining to various attributes range vary differently. Thus, in order to make them a bit more similar, we use the MinMaxScaler function from the preprocessing module of the sklearn library. In this way, we scale our data as per our requirement and prepare it for training.

## VI. METHODOLOGY

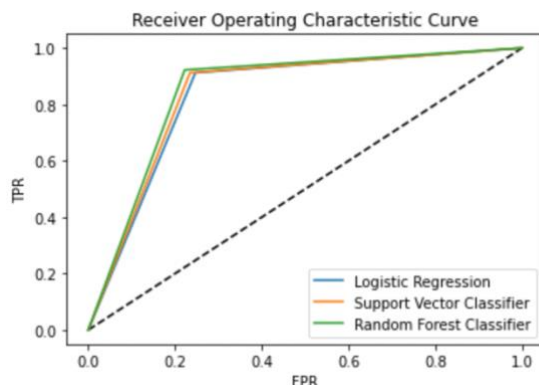
Once the data has been preprocessed and is ready for training, we move ahead with creating the models and training them. Initially, the logistic regression model is created. The 'liblinear' solver for Logistic Regression model is chosen as this is going to be a binary classification, heart disease or no heart disease. Similarly, the Support Vector Classifier of the Support Vector Machines is created and trained on the respective training dataset. Finally, the third model, Random Forest Classifier is instantiated. The number of estimators for the Random Forest Classifier is kept as 11, as 11 is the number of features in our dataset (excluding the column 'HeartDisease' as it is a target and also, as it cannot be considered a feature because it is supposed to be predicted as a result of the classifiers).

Usually, if the dataset is trained onto the entire dataset, we cannot test it anywhere and cannot make sure of any results it might give, and cannot make out how true those results would be without testing it. Once the model is trained on some data, it must be tested and evaluated on unseen data as an evaluation for how good the model performs on unseen data. Thus, it can be verified if the model is

actually learning to predict or is it just cramming up the trained data, which is also called as overfitting. So, we split the current dataset into two parts, train set and test set with a split of 0.8 respectively. The train dataset is then used to train all three classification models that have been instantiated by us. Once the models are trained, they'll be ready to be evaluated to test their accuracy and relevant metrics.

## VII. RESULTS AND CONCLUSION

Upon testing the various machine learning classification models, we get a variety of results. An accuracy of approximately 84.24% is achieved for the Logistic Regression model. Similarly, the Support Vector Classifier gives an output accuracy of 84.78%. On the other hand, the Random Forest Classifier provides the best accuracy among the three for almost 85.87%. An ROC curve can also be plotted for all three models in the same graph, which will then help to compare all three models.



*Figure 9. ROC Curve for all three models: Logistic Regression, Support Vector Classifier and Random Forest Classifier*

Also, comparison for various evaluation metrics can be done and the results for each model are as follows:

| Model               | Logistic Regression Classification |
|---------------------|------------------------------------|
| Accuracy            | 0.84239                            |
| F1 score            | 0.86635                            |
| Recall Score        | 0.91262                            |
| Mean Squared Error  | 0.15760                            |
| Mean Absolute Error | 0.15760                            |

| Model               | Support Vector Classification |
|---------------------|-------------------------------|
| Accuracy            | 0.84782                       |
| F1 score            | 0.87037                       |
| Recall Score        | 0.91262                       |
| Mean Squared Error  | 0.15217                       |
| Mean Absolute Error | 0.15217                       |

| Model               | Random Forest Classification |
|---------------------|------------------------------|
| Accuracy            | 0.85869                      |
| F1 score            | 0.87962                      |
| Recall Score        | 0.92233                      |
| Mean Squared Error  | 0.14130                      |
| Mean Absolute Error | 0.14130                      |

Thus, from all these observations and conclusions, it can be said that Random Forest Classifier worked the best among all three Machine Learning Classification Models namely, Logistic Regression, Support Vector Classification and Random Forest Classification. It has a better accuracy, a better F1 score, a better recall score and lowest mean squared error as well as mean absolute error among the three models.

This project can be even further developed with use of the state-of-the-art Deep Learning techniques and models. It would further be researched well if the research gets a touch of a medical background and testing is carried out in a realistic scenario, definitely not as a sole decision maker but just as a testing procedure so as to decide how good the model can be made with respect to a practical standpoint of view.

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